CS & IT ENGINEERING

Operating System

Synchronization

Lecture - 06

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Recap of Previous Lecture

Topic Questions on Semaphore

Topic Solution Without Busy Waiting

Topics to be Covered

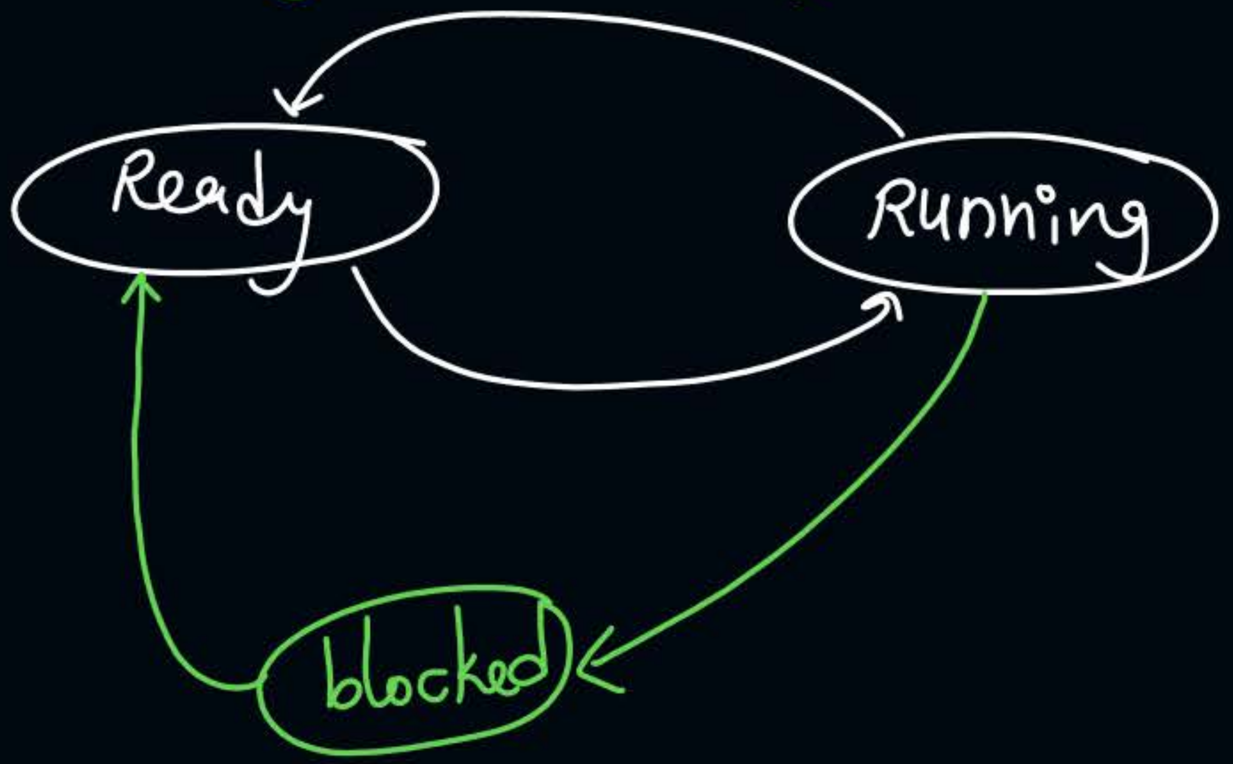
Topic Solution Without Busy Waiting

Topic Producer-Consumer Problem

Topic Reader-Writer Problem

Busy waiting :- (spin lock)

A process is busy in execution on CPU but still waiting for critical section. (Process is wasting CPU time)




```
while(condition);
```

Deadlock :-

A situation in which 2 are more processes can never execute further.

P1	P2
wait(S1)	wait(S2)
wait(S2)	wait(S1)

$S1 = \cancel{1}\ 0$

$S2 = \cancel{1}\ 0$

	P1	P2
CPU	wait(S1)	wait(S2) . . . . . .

P1, P2

$\Downarrow$

In deadlock

Topic : Solutions Without Busy Waiting

s = 1

```
wait(Semaphore s){
    s=s-1;
    if (s<0) {
        // add process to queue
        block();
    }
}
```

```
signal(Semaphore s){
    s=s+1;
    if (s<=0) {
        // remove
process p from queue
        wakeup(p);
    }
}
```

wait(s)
c. s.
signal(s)

s = ~~1~~ ~~0~~ ~~-1~~ ~~-2~~
~~-1~~ ~~0~~ 1

p1 → CS
p2 ⇒ ✓
p3 ⇒ ✓

Queue

~~p2~~	~~p3~~	

blocked

Topic : Classical Problems of Synchronization

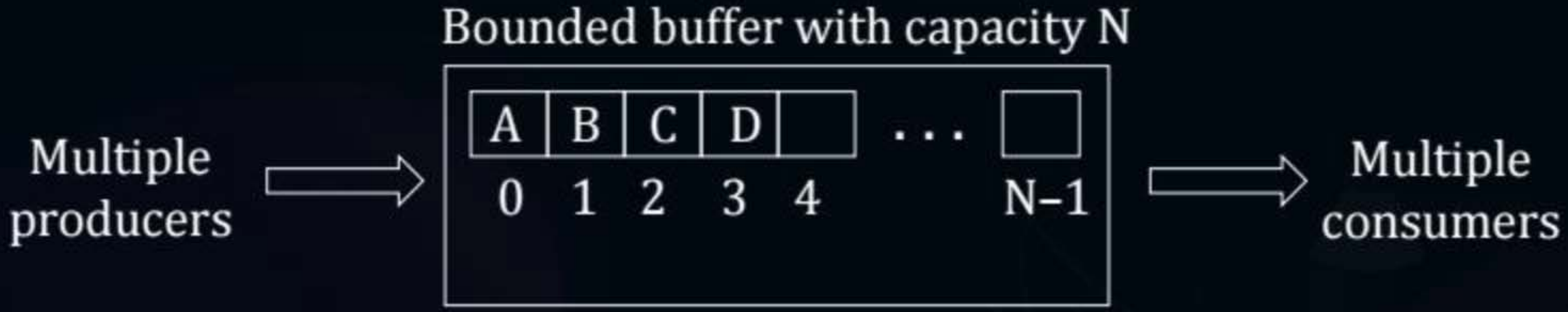
→ Producer-Consumer (Bounded-buffer)

→ Reader- writer

→ Dinining philosopher

Topic : Bounded Buffer Problem

array



Topic : Bounded Buffer Problem

- Known as producer-consumer problem also
- Buffer is the shared resource between producers and consumers

Topic : Bounded Buffer Problem : Solution

semaphore values

- Producers must block if the buffer is full ⇒ Full = N. Empty = 0.
- Consumers must block if the buffer is empty

→ Full = 0
Empty = N

Topic : Bounded Buffer Problem : Solution

- **Variables:**
  - Mutex: Binary Semaphore to take lock on buffer (Mutual Exclusion)
  - Full: Counting Semaphore to denote the number of occupied slots in buffer
  - Empty: Counting Semaphore to denote the number of empty slots in buffer

Initialization :-

Mutex = 1

Full = 0

Empty = N

Topic : Producer() & Consumer() Processes

```
wait(Empty)
wait(Mutex)
//add item on buffer
signal(Mutex)
signal(Full)
```

```
wait(Full)
wait(Mutex)
// remove item from buffer
signal(Mutex)
signal(Empty)
```

Note:- If first 2 wait() statements are swapped in producer then there can be deadlock when buffer is full.

Note:- If first 2 wait() statements are swapped in Consumer then there can be deadlock when buffer is Empty.

Reader-writer problem:-

A file is shared b/w multiple reader & writer processes.

	allowed	
currently	Read	write
write	✗	✗
Read	✓	✗

Topic : Reader-Writer Problem Solution

- If writer is accessing the file, then all other readers and writers will be blocked
- If any reader is reading, then other readers can read but writer will be blocked

2 mins Summary

Topic Solution Without Busy Waiting

Topic Producer-Consumer Problem

Topic Reader-Writer Problem

Happy Learning

THANK - YOU